The Solutions for EE3.18 and AO12, 2025

Model answer to Q 1(a): Bookwork and Discussions in Class

i) Signal/noise ratio becomes more problematic as frequency decreases below 1 GHz, because of the increase in galactic noise. As a result, more signal power is required to maintain a good signal/noise ratio.

[4]

ii) The frequency between 1 GHz and 10 GHz is convenient for commercial exploitation because both galactic and atmospheric noise are very low. Also, cheap lumped-element passive components and high gain active components can be used in this frequency band.

[4]

iii) For microwave ovens, high power has to be generated at low cost. At higher frequencies it becomes more difficult to generate high power and also increased tolerances increase the cost of manufacturing. Moreover, cooking skin depth decreases and, therefore, it becomes more difficult to cook uniformly. At lower frequencies, there will be fewer hot-spots, and these will be spaced further apart, again making it difficult to cook uniformly. Moreover, the size and weight of the magnetron would increase.

[4]

iv) The water absorption peaks are found at 22 GHz and 183 GHz; while the oxygen absorption peaks are found at 60 GHz and 119 GHz.

[4]

v) At 38 GHz and 94 GHz there are troughs in the atmospheric attenuation curves. As a result, Line-of-Sight communication links (having wide bandwidths) and directional radar systems (having highly directional antennas) are used in these bands, respectively.

[4]

vi) The 60 GHz band has very high atmospheric attenuation. As a result, it is ideal for indoor wireless local area network applications and inter-satellite communication links. The potential interference to other nearby services operating in the same band can be minimized by the inherent drop in signal strength, as distance increases.

[4]

Model answer to Q 1(b): Discussions in Class and New Calculations

i) Unloaded Q-factor for a tuned circuit can be represented by that of the inductor, by assuming all the loss resistance is effectively lumped into the inductor: $Q = \omega L / R$, where ω = angular frequency; L = inductance and R = equivalent loss resistance.

[4]

ii) The simple transition cut-off frequency for a field-effect transistor is given by: $f_T = \frac{g_{mo}}{2\pi C_{gs}} = 32.6\,GHz$. No, the unity-gain bandwidth is defined by the maximum frequency of oscillation, f_{max}.

[5]

iii) With a suitable inductor or capacitor and bias network, a transistor can be configured to create a negative resistance that can cancel out the effective loss resistance in a tuned circuit. This “active resonator” technique is possible as long as the operating frequency is at least an order of magnitude lower in frequency to that of the transition cut-off frequency of the transistor.

[4]

Answers

Model answer to Q 1(c): Bookwork and Discussions in Class

i) The simple relationship between electrical length and frequency for a transmission line operating in the (quasi-)TEM mode is: $\theta = \frac{\omega\sqrt{\varepsilon_{r\ effective}}}{c} l$, where c = speed of light in a vacuum, l = physical length and $\varepsilon_{r\ effective}$ = effective relative permittivity.

[5]

ii) Integrated circuits employ transmission lines of fixed electrical length. Therefore, as frequency decreases the lengths of line increase. This can be a problem, as the large size may be prohibitive in terms of space available and/or cost. As frequency increases, sensitivity to the tolerances of open-circuit stubs also increases, but chip package density can increase.

[4]

iii) In general, the attenuation constant of an air-filled coaxial cable is directly proportional to the surface resistance, given by: $Rs = \sqrt{\frac{\omega\mu}{2\sigma_o}}$, where μ = permeability of the metal conductor and σ_o = bulk DC conductivity of the metal conductor. As a result, attenuation is directly proportional to the square root of frequency.

[4]

Model answer to Q2: New Calculations

Perspex, having a relative permittivity $\varepsilon_r = 2.6 - j\,0.018$ at 10 GHz, is a low loss dielectric, which is non-magnetic and isotropic. The propagation constant γ and intrinsic impedance η can be determined by the following expressions:

$$\gamma \cong 2\pi k\left(0.5\tan\delta + j\right) \quad and \quad \eta = \frac{j\omega\mu}{\gamma}$$

where k, $\tan\delta$, ω, and μ have their usual meanings

(i) $$\tan\delta = \frac{\varepsilon''_r}{\varepsilon'_r} = 6.9x10^{-3}$$
$$\therefore Q = \frac{1}{\tan\delta} = 145$$

[3]

(ii) $$\lambda = \frac{v_p}{f} = \frac{c}{f\sqrt{\varepsilon'_r}} = 18.6\ [mm]$$
$$k = \frac{1}{\lambda} = 53.75\ [\lambda/m]$$
$$\therefore \gamma \approx 1.165\ [Np/m] + j337.7\ [rad/m]$$

[4]

(iii) $$\delta = \frac{1}{\alpha} = 858\ [mm]$$

[3]

(iv) $$l = \lambda/4 = 4.65\ [mm]$$

[3]

Answers

(v) $Power\ Attenuation = e^{-2(\alpha l)}$
$\therefore \alpha l = 0.0217\ [Np/\lambda]$
$\therefore Power\ Attenuation = \alpha l(20\ log\ e) = 8.686\alpha l = 0.188\ [dB/\lambda]$

[4]

(vi) $\eta = \frac{j\omega\mu}{\gamma} \quad where \quad \omega = 2\pi f \quad and \quad \mu = 4\pi x10^{-7}\ [H/m]$
$\therefore \eta = 233.8 + j0.8\ [\Omega]$

[4]

(vii) $P_D = \frac{|E|^2}{Re\{\eta\}} = 4.3\ [mW/m^2]$

[4]

Model answer to Q3(a): Discussion in Class

(i) The velocity factor is the ratio of the speed of waves on the transmission line to the speed of light, *c*, in vacuum, where *c* = 3x10^8 m/s. The velocity factor is a dimensionless number less that unity. It basically represents how much the wave is slowed down inside the transmission line, when compared to propagation in free space.

[3]

(ii) The characteristic impedance is the intrinsic impedance for an infinitely long length of transmission line, or of a finite length of line at times before any reflections have arrived back at the measuring source. It is important to match two lengths of different propagating media to the same impedance, in order to avoid reflected waves and associated reflected power. Conversely, two specific impedances are required to generate a specific reflected wave.

[3]

(iii) The return loss is the number of dB by which the return wave power is less than the forward incident wave power. This is important because it gives a quantitative measure of how much reflected power is generated and thus indicates the level of impedance mismatch.

[3]

(iv) The reflection coefficient is the complex ratio of reflected wave amplitude to incident wave amplitude. This is important, as it allows the exact impedance at a discontinuity to be calculated.

[3]

(v) The phase delay is the amount of phase shift experienced by the wave in travelling along a certain length of transmission line. It is important to be able to calculate this delay so that any reflected or transmitted waves can be combined; for example, in-phase or 180° out-of-phase.

[3]

Model answer to Q3(b): New Calculations

$$Zo = \sqrt{\frac{L}{C}}; \quad v_p = \frac{1}{\sqrt{LC}} \quad and \quad VF = \frac{v_p}{c}$$

[2]

If *Zo = 50* Ω and *VF = 0.6*, *L = 87.8 nH* and *C = 351 pF* for a length of 1 m transmission line.

[2]

Answers

Model answer to Q3(c): New Calculations

$$z_L = \frac{Z_L}{Zo} = 1.5 + j0.26\Omega$$

$$\rho = \frac{z_L - 1}{z_L + 1} = 0.21 + j0.08 \equiv 0.22e^{+j21.54}$$

$$\text{Return Loss} = 13\ dB$$

[2]

As the load impedance is moved towards the input reference plane, the modulus of the complex reflection coefficient stays the same (assuming the line is lossless), but the phase angle is increased by 180° for each λg/4 reduction of the length of the interconnected transmission line.

[4]

Model answer to Q4: Bookwork and New Calculations

(i) $\gamma_c = jk_c = j\frac{2\pi}{\lambda_c} = j\frac{\pi}{a}$ since $\lambda_c = 2a$ for the TE_{10} mode

[3]

(ii) $\gamma_0 = jk_0 = j\frac{2\pi}{\lambda_0} = j\frac{\omega_0}{c}$ since $\lambda_0 = \frac{c}{f_0}$

[2]

(iii) When $f_0 < f_c$:

$\lambda_0 > \lambda_c$ and $k_0 < k_c$ $\therefore |\gamma_0| < |\gamma_c|$ and $f_0 < f_c = \frac{c}{2a}$

$\therefore \gamma_g = \sqrt{k_c^2 - k_0^2} \equiv \alpha + j0$ and $\alpha = \sqrt{\left(\frac{\pi}{a}\right)^2 - \left(\frac{\omega_0}{c}\right)^2}$

i.e. Evanescent (non-propagating) Mode, because there is no phase constant

[5]

(iv) When $f_0 > f_c$:

$\lambda_0 < \lambda_c$ and $k_0 > k_c$ $\therefore |\gamma_0| > |\gamma_c|$ and $f_0 > f_c = \frac{c}{2a}$

$\therefore \gamma_g = \sqrt{k_c^2 - k_0^2} \equiv 0 + j\beta$ and $j\beta = \sqrt{\left(\frac{\pi}{a}\right)^2 - \left(\frac{\omega_0}{c}\right)^2}$

i.e. Propagating Mode, because there is a phase constant

[5]

(v) With $a = 22.86$ mm, $\lambda_c = 2a = 45.72$ mm and $f_c = \frac{c}{\lambda_c} = 6.562$ GHz

When $f_0 = f_c - 1\ \mathrm{GHz} = 5.562\ \mathrm{GHz}$, $\gamma_g = \alpha = \sqrt{\left(\frac{\pi}{a}\right)^2 - \left(\frac{\omega_0}{c}\right)^2} = \sqrt{18{,}886 - 13{,}570} = 72.9\ \mathrm{Np/m}$

When $f_0 = f_c + 1\ \mathrm{GHz} = 7.562\ \mathrm{GHz}$, $\gamma_g = j\beta = \sqrt{\left(\frac{\pi}{a}\right)^2 - \left(\frac{\omega_0}{c}\right)^2} = \sqrt{18{,}886 - 25{,}084} = j78.7\ \mathrm{rad/m}$

[10]

Answers